

Journal Pre-proofs

Atmospheric concentrations of CH₄ in central-western areas of Brazil for 2009-2019 using GOSAT satellite

Luciano de Souza Maria, Fernando Saragosa Rossi, Luis Miguel da Costa, Marcelo Odorizzi Campos, Alan Rodrigo Panosso, Carlos Antonio da Silva Junior, Newton La Scala Jr

PII: S0273-1177(25)00089-4
DOI: <https://doi.org/10.1016/j.asr.2025.01.058>
Reference: JASR 18126



To appear in: *Advances in Space Research*

Received Date: 28 July 2024
Revised Date: 22 January 2025
Accepted Date: 23 January 2025

Please cite this article as: Maria, L.d.S., Rossi, F.S., da Costa, L.M., Campos, M.O., Panosso, A.R., da Silva Junior, C.A., Scala, N.L. Jr, Atmospheric concentrations of CH₄ in central-western areas of Brazil for 2009-2019 using GOSAT satellite, *Advances in Space Research* (2025), doi: <https://doi.org/10.1016/j.asr.2025.01.058>

This is a PDF file of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability, but it is not yet the definitive version of record. This version will undergo additional copyediting, typesetting and review before it is published in its final form, but we are providing this version to give early visibility of the article. Please note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2025 COSPAR. Published by Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Atmospheric concentrations of CH₄ in central-western areas of Brazil for 2009-2019 using GOSAT satellite

Luciano de Souza Maria¹; Fernando Saragosa Rossi¹; Luis Miguel da Costa¹; Marcelo Odorizzi Campos¹; Alan Rodrigo Panosso¹; Carlos Antonio da Silva Junior²; Newton La Scala Jr¹

¹São Paulo State University (Unesp), Jaboticabal 14884-900, Brazil.

² University of the State of Mato Grosso, Sinop 78580-000, Brazil.

Abstract

The study of methane (CH₄) concentrations in the atmosphere over central-western Brazil is of great importance due to the significant impact of cattle-raising activities and the presence of one of the largest floodplains in the world, the Pantanal. Furthermore, an aggravating factor observed in recent years is the prolonged dry periods, which have intensified the occurrence of fires. These fires, whether in deforested areas or regions converted for livestock or agricultural use, contribute to the rising atmospheric concentrations of CH₄. We analyzed the seasonality and spatial variability of the column-averaged CH₄ dry air mole fraction (xCH₄) and investigated its relationships with cattle ranching activities and other attributes derived from remote sensing in the State of Mato Grosso do Sul. We used the Enhanced Vegetation Index (EVI) and Land Surface Temperature (LST) data from the MODIS sensor, and the xCH₄ data was acquired from the TANSO-FTS sensor onboard on GOSAT. The Mann-Kendall trend test revealed a significant trend ($P < 0.05$) for April, suggesting a monotonic increase in xCH₄ levels after 2014, further confirmed by Pettitt's test ($P = 0.042$). Moreover, we were able to explain 78.43% of the data variability through PCA analysis, and the first component alone was responsible for 43.42%. The xCH₄ pattern was highly similar to the LST over the time series (2009-2019). The linear regression between both points out that xCH₄ levels would increase around 3.57 per °C. Therefore, with climate change, the temperature tends to be an excellent indicator of the increase in CH₄ concentration in areas located in the central-west region of Brazil.

Keywords: greenhouse gases, methane, radiative forcing, climate change, cattle heads, remote sensing

1. Introduction

Mato Grosso do Sul (MS), located in the central-western region of Brazil, covers an area of approximately 357 thousand km² (IBGE, 2020), and livestock represents an important agricultural activity and plays an important economic role (Bungenstab and Gomes, 2016). Similarly, the Pantanal, one of South America's largest wetlands, relies on cattle ranching as its primary economic activity (Bergier et al., 2019b). Moreover, the state of MS represents emerging frontiers for the expansion of bioenergy crops, particularly sugarcane cultivation (Walter et al., 2021). In recent decades, these areas have drawn significant attention due to deforestation and enteric methane emissions from cattle herds. Consequently, the MS ranked as the fifth-largest emitter of greenhouse gases (GHG) (46 million CO₂eq) in Brazil in 2020, primarily driven by

emissions from the agricultural sector, including livestock and the use of synthetic fertilizers (SEEG, 2023).

Anthropogenic emissions of carbon dioxide (CO₂) and methane (CH₄) are major drivers of radiative forcing, contributing significantly to the current climate crisis. In 2019, the net change in Earth's energy balance was measured at 2.72 W m⁻², reflecting a sharp increase since the 1970s (IPCC, 2021). Since 1850 an increase of 1.09 °C in the global mean surface temperature has been observed, especially in recent years (Wang et al., 2021). There has been an exponential growth in CH₄ concentrations in recent decades (Kenea et al., 2021). For example, in 2023, the atmospheric CH₄ concentration reached 1921.88 ± 0.62 parts per billion (Lan et al., 2023), while global mean concentrations of CH₄ during the preindustrial revolution period varied only between 625 and 807 ppb (Gulev et al., 2023).

Natural methane (CH₄) emissions from tropical wetlands rank among the most significant global sources of atmospheric CH₄ (Bergier et al., 2019a). In MS, the Pantanal, which accounts for 25% of the state's territory (IBGE, 2020), has experienced extensive land-use changes driven by the expansion of cattle ranching (Teodoro et al., 2022). In addition, these tropical wetlands are highly sensitive to environmental factors, including temperature (Bansal et al., 2018) and precipitation patterns (Parker et al., 2016). Climate anomalies, such as the El Niño-Southern Oscillation (ENSO) and shifts in Atlantic Sea Surface Temperature (SST), further amplify CH₄ emissions, underscoring the critical interplay between anthropogenic and natural drivers (Barbosa et al., 2019).

Anthropogenic activities are responsible for approximately 60% of global methane (CH₄) emissions, with the primary sources being fossil fuel combustion, livestock production, and biomass burning (Saunois et al., 2020; Tian et al., 2016, 2015). Agropastoral systems, particularly rice cultivation and livestock, are among the most significant contributors to these emissions (Dangal et al., 2017; Wu et al., 2017). Between 2008 and 2017, the livestock sector alone emitted an average of 111 Tg CH₄ per year, accounting for nearly 30% of global anthropogenic CH₄ emissions (Saunois et al., 2020). In Brazil, methane emissions linked to animal agriculture are predominantly driven by enteric fermentation in ruminants, land-use and land-cover changes (LULC), agricultural residue burning, and waste management (Mendonça et al., 2020). In 2015, Brazil's total CH₄ emissions were 20.1 Tg, with enteric fermentation contributing a significant 71% of this total (Crippa et al., 2020).

Despite considerable progress in the field, critical knowledge gaps remain regarding the temporal patterns of atmospheric CH₄ emissions from production systems and the complex interactions between anthropogenic emissions and climate change (Mousavi and Falahatkar, 2020). In this scenario, the use of remote sensing provides an effective way to monitor climate seasonality and GHG emitting sources at regional and global scales (Chen et al., 2021; Mustafa et al., 2021). However, there are still many gaps in temporal trends of atmospheric CH₄ emissions in production systems, and the relationship between anthropogenic emissions, activities and climate change (Mousavi and Falahatkar, 2020).

Currently, several satellites monitor GHG emissions, such as Greenhouse gases Observing SATellite (GOSAT) satellite of the Japan Aerospace Agency (JAXA) (Kuze et al., 2009) and the Sentinel-5 satellite (ESA), AIRS (Atmospheric Infrared Sounder),

and SCIAMACHY (Scanning Absorption Spectrometer for Atmospheric Cartography) satellite sensors. The sensors onboard these satellites use the xCH_4 (Mousavi et al., 2017). They detect the CH_4 concentration for the mid and upper troposphere in the thermal infrared (TIR) region and the region closest to the surface in the shortwave infrared (SWIR) regions (Kuze et al., 2009).

Since 2009, the GOSAT satellite has provided consistent global coverage of atmospheric CH_4 , offering invaluable insights into long-term trends through extensive ten-year time series data, surpassing other available orbital datasets in both resolution and scope (Ross et al., 2013; Wilson et al., 2020). Accordingly, the objectives of this study were: (i) to assess CH_4 trends using remote sensing tools across a time series from 2009 to 2019; (ii) to verify the existence of xCH_4 trends with variables obtained by remote sensing; (iii) to observe whether xCH_4 trends relate to the increment of the cattle herd in the studied areas (Figure 1).

2. Material and Methods

2.1 Study area

The state of Mato Grosso do Sul, located in the Midwest region of Brazil (Fig. 1), encompasses a vast and ecologically diverse territory characterized by tropical and subtropical climates. Average air temperatures range between 24 °C and 40 °C, and the region receives an average annual precipitation of approximately 1,500 mm (Alvares et al., 2013; Beck et al., 2018). Covering a total area of 357,147.99 km² (IBGE, 2020), the state hosts a rich array of natural resources distributed across three major biomes: the Cerrado, the Atlantic Forest, and the Pantanal (Teodoro et al., 2016).

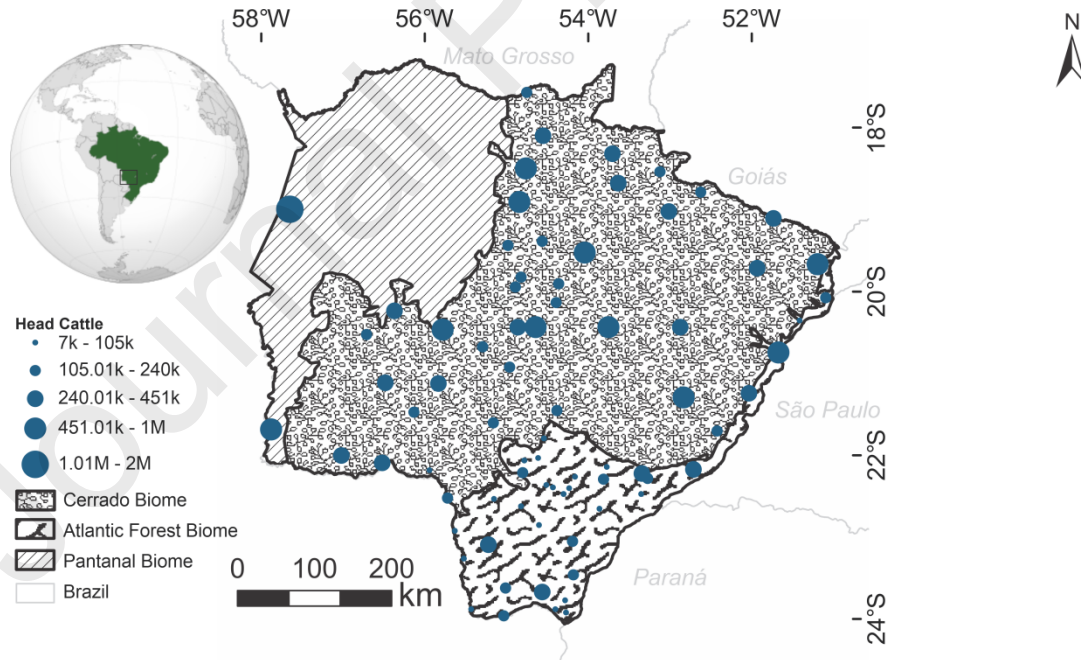


Fig. 1. Map of the State of Mato Grosso do Sul with the average EVI of the year 2019 and the annual sum of the cattle heads.

The Cerrado biome is characterized by a wide range of phytophysionomies, including pastures, “Cerradão”, “Veredas”, and “Riparian forests” and is the second largest biome in Brazil, covering approximately 22% of Brazil's territory (Abreu et al., 2020). The predominant soil types in the Cerrado are Ferralsols (40%), Arenosols (16%), Acrisols (~12%), and Plinthosols (~10%) (Gomes et al., 2019).

The Atlantic Forest is recognized as a global biodiversity hotspot due to its exceptional species richness; however, it is among the most degraded biomes in South America, with only ~8% of its original coverage remaining (de Lima et al., 2020; Morellato and Haddad, 2000; Teodoro et al., 2016). The dominant soil classes in this biome include Ferralsols (~37%), Acrisols (~30%), and Cambisols (~15%).

The Pantanal biome, occupying approximately 25% of southern Mato Grosso do Sul, is defined by extensive seasonal flooding, creating unique ecological conditions. The predominant soil types in the Pantanal are Planosols (~35%), Plinthosols (~20%), and Podzols (~20%).

2.3. Precipitation Data – CHIRPS

Precipitation data utilized in this study were derived from the Climate Hazards Group Infrared Precipitation with Stations (CHIRPS). This dataset, developed collaboratively by the United States Geological Survey (USGS) and the Climate Hazards Group at the University of California, Santa Barbara (UCSB), provides high-resolution precipitation records. CHIRPS products are freely accessible via the official website (<https://data.chc.ucsb.edu/products/CHIRPS-2.0/>) and offer extensive spatial coverage, ranging from 50° N to 50° S latitude, with a spatial resolution of 0.05° (~5 km). Temporal resolution includes daily, pentadal, and monthly aggregation, making it a valuable resource for climatological analyses.

The latest version, CHIRPS 2.0, integrates precipitation data from over 11,000 stations in Brazil alone (Marengo et al., 2021), and provides records spanning from 1981 to the present. For this study, CHIRPS data were retrieved for the State of Mato Grosso do Sul for the period 2009–2019 using the Google Earth Engine platform. Data were accessed in the ee.ImageCollection format ("UCSB-CHG/CHIRPS/DAILY") through JavaScript code to acquire images in TIFF format (Funk et al., 2015). Annual precipitation sums were calculated for each pixel within the time series, enabling a spatially explicit assessment of precipitation trends.

2.4. Land Surface Temperature Data

Daytime and nighttime data from the MOD11A2 V006 product (Wan, Z., S. Hook, 2015) produced and made available by NASA (National Aeronautics and Space Administration), from the MODIS (Moderate-Resolution Imaging Spectroradiometer) sensor with 8 days with a spatial resolution of 1 km². Each pixel value in MOD11A2 is a simple average of all corresponding MOD11A1 LST pixels collected within 8 days, as this is the exact revisit period of the Terra and Aqua satellites derived from the MODIS sensor.

The annual average of the MOD11A2 product data was calculated using the Raster Calculator tool of the QGIS 3.16.4-Hannover software. Subsequently, the results of the surface temperature calculation without units were multiplied by the scale factor of 0.02

to obtain degrees Kelvin (K), then converted to Celsius (°C) (equation A) and divided by 8 to obtain the daily values.

$$LST_{Celsius_{Day}} = \frac{LST_{Day_{1km}} \cdot 0.02 - 273.25}{8} \quad \text{Eq. (A)}$$

2.5. Enhanced Vegetation Index data

The Enhanced Vegetation Index (EVI) proposed by Huete et al. (1997), besides being more sensitive in dense vegetation areas, this index also corrects for some atmospheric conditions and canopy background noise. It incorporates an “L” value to adjust for canopy background, “C” values as coefficients for atmospheric resistance, and values from the blue band (B) while reducing the background noise, atmospheric noise, and saturation in most cases (Vermote et al., 2016). The values can vary between -1 and 1, in which vegetation is indicated by positive values, as there is greater reflectance in the red-light wavelength than in the NIR wavelength (Huete et al., 2002). For this study, we used the MOD13Q1 product (MODIS sensor), of the Earth orbital platform with a spatial resolution of 250 m was composed in Google Earth Engine derived from for the period from 2009 to 2019.

$$EVI = g \frac{\rho_{IVP} - \rho_V}{\rho_{IVP} + (c1 * \rho_V) - (c2 * \rho_A) + L} \quad \text{Eq. (B)}$$

Where: ρ_{IVP} , ρ_V , and ρ_A are the reflectances in the near-infrared, red, and blue spectral ranges, respectively. g is the gain factor (2.5), $c1$ and $c2$ are the correction coefficients for atmospheric red (6) and blue (7.5) effects, respectively, and L is the correction factor for ground interference (1).

2.6. Cattle Heads Data

Annual data on the Brazilian cattle herd were obtained from the Brazilian Institute of Geography and Statistics (IBGE, 2020). These data, measured at the end of each year, represent annual totals and were analyzed at the municipal level for the study area. For the State of Mato Grosso do Sul (MS), the total number of cattle was calculated by aggregating the livestock counts from all municipalities for each year in the time series (2009–2019).

The analysis revealed that, in 2019, the municipality of Corumbá ranked second nationally among Brazilian cities, with 1.8 million head of cattle. On a broader scale, Mato Grosso do Sul was the fifth most significant state in terms of cattle population, with a total of 19.4 million head (IBGE, 2020). These results underscore the importance of MS as a major contributor to Brazil’s cattle industry during the study period.

2.7. Average Column Data of CH₄ (xCH₄)

The GOSAT satellite was launched in January 2009 into a helio-synchronous orbit with a local equator crossing time of 13:00 hours. Observations in standard mode are made in three circular pixels over a cross-section of 790 km with a spatial resolution of 10.5 km. It is worth noting that in standard mode, the pixels are separated by approximately 263 km along the swath, which results in temporal resolution of coverage every 3 days. The Thermal and Near Infrared Sensors for Carbon Observing Fourier Transform Spectrometer (TANSO - FTS) and a Cloud and Aerosol Imager (TANSO-CAI), made it

possible to retrieve the atmospheric xCH_4 column by solar backscatter nadir measurements, with three channels in the shortwave infrared (0.758 - 0.775 μm , 1.56 - 1.72 μm , and 1.92 - 2.08 μm) and another channel in the thermal infrared region (5.5-14.3 μm) (Kuze et al., 2009).

2.8. Digital processing

The steps from data acquisition to the generation of the result were as follows: 1. The acquisition of data through the University of Leicester GOSAT Proxy version 9.0 in .nc file format (Parker, R.; Boesch, 2020). These files are in a biweekly temporal aggregation for the entire world comprehending the period of April 2009 to December 2019 (https://data.ceda.ac.uk/neodc/gosat/data/ch4/nceov1.0/CH4_GOS_OCPR/) and processed files are available at GitHub (<https://github.com/TioLucio/CEDA/tree/master>). 2. The entire process of filtering the data, filtering outliers and then aggregating annual and monthly averages for each coordinate was performed in the R programming language. 3. the formal analysis, with the descriptive statistic, Grey Relation analysis (GRA), Principal Component Analysis and Mann–Kendall test.

We used the regression method (Gujarati and Potter, 2011) to remove the long-term trend from the xCH_4 dataset, isolating natural and regional variability. This approach involved subtracting modeled trends, minimizing the influence of atmospheric changes over time. The xCH_4 data, validated over the Amazon biome using aircraft profiles and GOSAT orbital data (Webb et al., 2016), were analyzed using GOSAT geospatial coordinates (10.5 km pixel resolution, JAXA) processed in QGIS. This ensured consistency across temporal resolutions and orbital sensor data.

For the dry air column average CH_4 mole fractions (xCH_4) were derived using the proxy method of using the atmospheric transport model of CO_2 , thus the method multiplies the ratio xCH_4/xCO_2 by a model of in situ xCO_2 observations (Parker et al., 2011; 2015). The xCO_2 model was based on local measurements of land surfaces using the median of the global models: GEOS-Chem (Feng et al., 2011), Carbon Tracker (Peters et al., 2007), and LMDZ (Chevallier et al., 2010). The xCH_4 Proxy data shows an overall global bias of 9.06 ppb (Parker et al., 2020).

2.9 Statistical Analysis

We use the Mann–Kendall ($P < 0.05$) test to determine the time series (2009 to 2019) statistical significance (Kendall, 1948). Furthermore, we use the Pettitt ($P < 0.05$) homogeneity test to detect the abrupt changes in the time series (2009 to 2019) (Pettitt, 1979). It is important to highlight that these non-parametric tests do not require a hypothesis in the data distribution. We also measure the similarity between variables with Grey Relation analysis (GRA), this quantitative analysis technique possible to evaluate the correlation between variables by relational degree (Deng, 1982).

We detect an association between xCH_4 and variables (LST, EVI, and Rainfall) and the total number of Cattle Heads with the Principal Component Analysis (PCA). PCA allows the reduction in the number of factors of new synthetic factors obtained by a linear combination of the original data, aiming to obtain the largest explained variance. The linear regression analysis was performed between the variables that provided some correlation with the xCH_4 because it presented analytical estimates of confidence

intervals, and robustness to outliers in the seasonal data, with by Levene test (Fig S1). Moreover, all analyses were performed with statistical software R version 4.1.2, through the libraries (ggplot2, trend, FactoMineR, factoextra, corrplot), due to its free access and being open source democratizes the statistical analyses (R Development Core Team, 2017).

3. Results

The monthly distribution of xCH_4 data over the time series from 2009 to 2019, we observed for June, July, and August that the average were smaller and with reduced amplitude compared to the other months (Fig. 2). For December, January, February, and March we observed the largest amplitudes and the highest averages values. The mean annual values of xCH_4 were higher than 1757.97 ppb, indicating a distribution of greater presence in the third quartile. The orbital data of xCH_4 exhibited monthly fluctuations in the time series from 2009 to 2019 (Fig. 2).

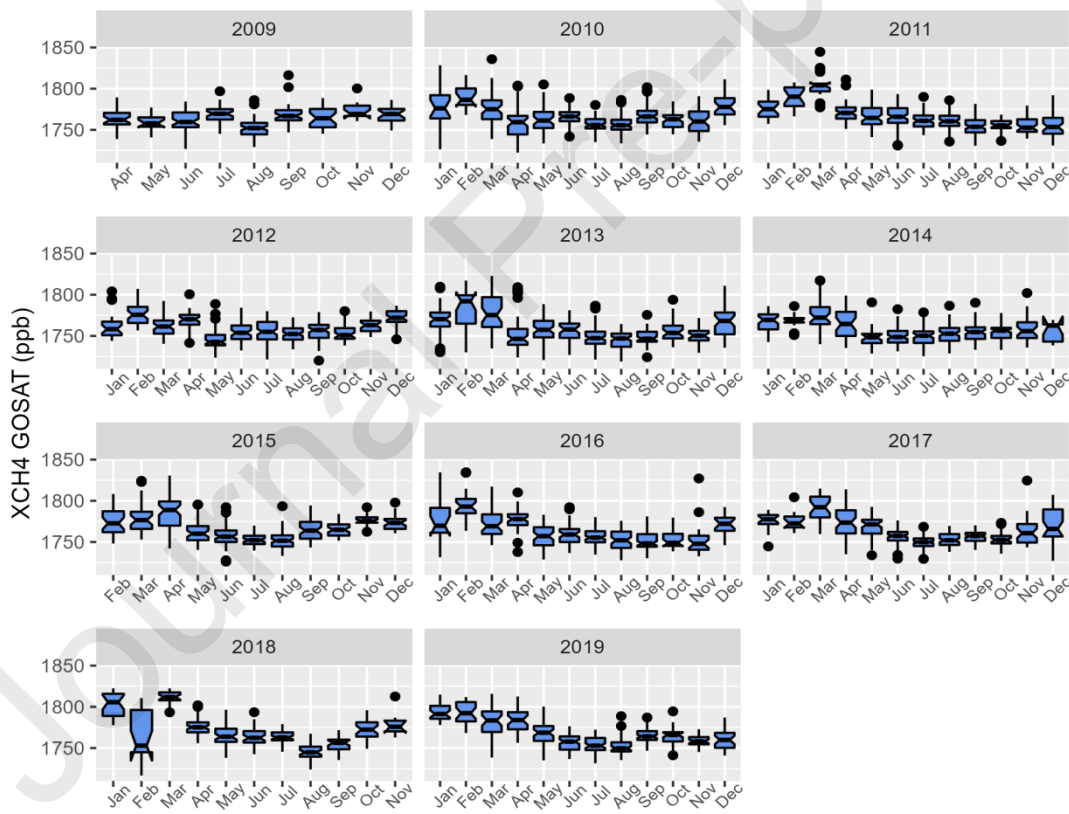


Fig. 2. Boxplot of atmospheric column monthly CH_4 (xCH_4) distribution for time series (2009-2019) in Mato Grosso do Sul, Brazil. xCH_4 data obtained from the TANSO FTS sensor (GOSAT Proxy) from the University of Leicester (Parker, R.; Boesch, 2020). Data from orbital observations of the variable was 668 (2009), 1464 (2010), 1135 (2011), 1223 (2012), 890 (2013), 726 (2014), 971 (2015), 1708 (2016), 1172 (2017), 1063 (2018) and 761 (2019).

Grey Relational Correlation Coefficient was possible to understand the similarity between the spatial distribution of the variables (Fig. 3). In this way, observed that most of the correlations between the variables range from 0.56 to 0.82. The EVI showed high similarity ($P < 0.82$) with the Rainfall. Likewise, the LST showed also high similarity ($P < 0.76$) with the variable XCH_4 . For the other variables, the grey correlation coefficients of the other indexes are lower than 0.75, which is at a relatively high level. (Fig. 3).

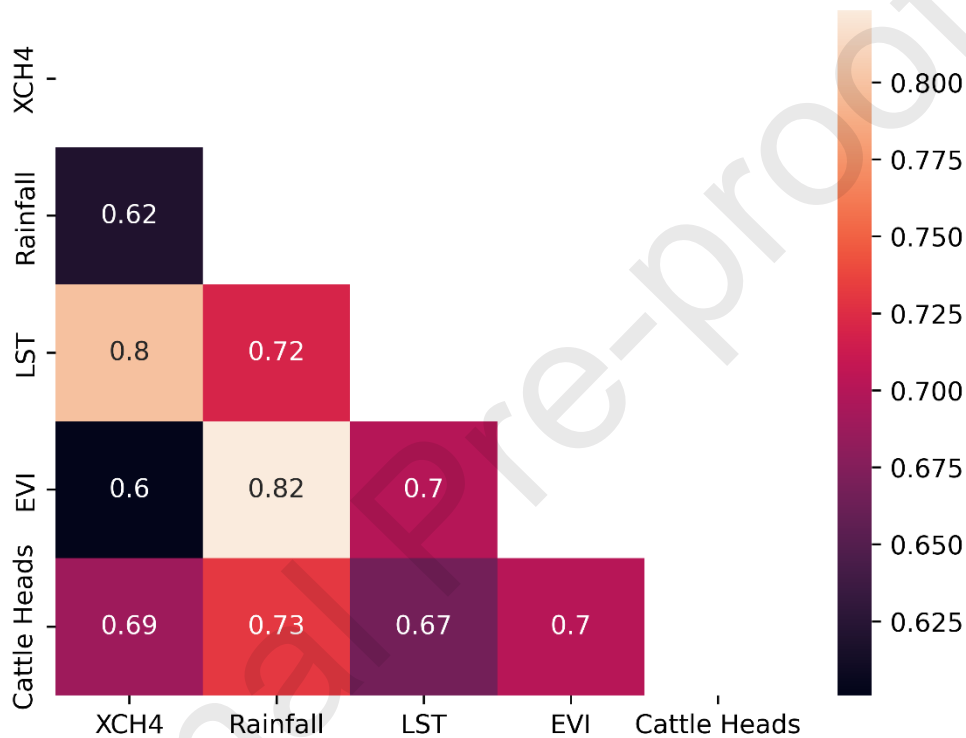


Fig. 3. Grey Relational Correlation Coefficient matrix between the XCH_4 and the Moderate-Resolution Imaging Spectroradiometer sensor products (Land Surface Temperature (LST) and Enhanced Vegetation Index (EVI)), Rainfall (CHIRPS) and Cattle Heads.

No upward trend was observed in the distribution of variable xCH_4 in the time series. The Mann-Kendall test was not significant at a 5% probability level ($P = 0.87$), statistically corroborating the confirmation of the absence of a monotonic increasing or decreasing pattern in the annual mean of xCH_4 . Thus, the Pettit test was not performed.

However, the monthly averages of xCH_4 in the time series showed values above the median for January, February, and March. Whereas, the other months provided values below or very close to the median (Fig. S2). Mann-Kendall test was significant at a 5% probability level ($P = 0.042$) indicating the presence of a trend for April throughout the time series and the Pettit test was significant at a 5% probability level ($P = 0.048$) indicating a monotonic growth after the year 2014. For the other months, this trend was not observed (Fig S2). Thus, it evidenced a growth of xCH_4 and LST for the period in the time series (Fig. 4).

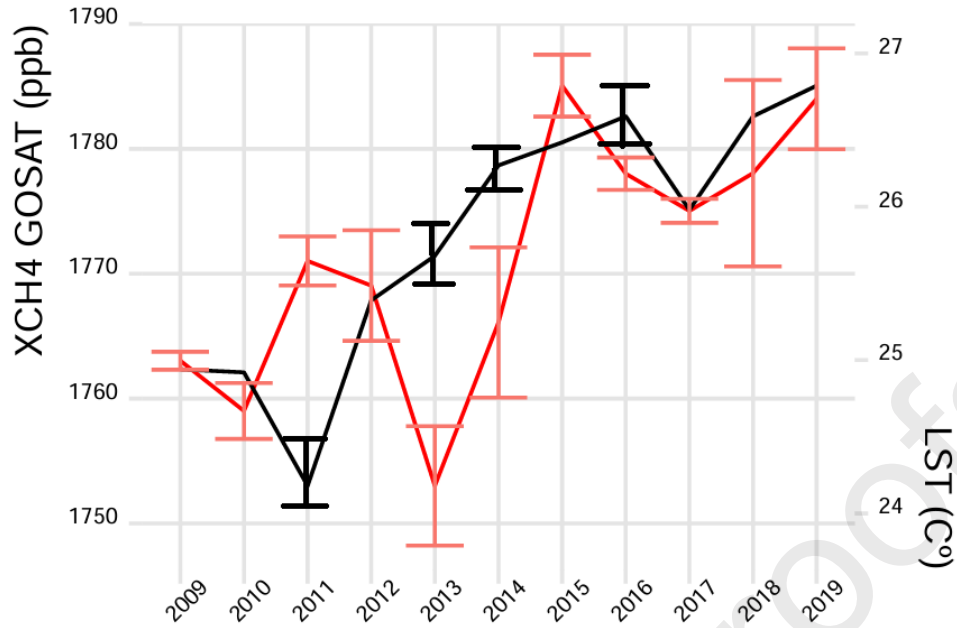


Fig. 4. The variations of Land Surface Temperature (LST °C) in the black lines and xCH₄ GOSAT in the red lines in April. The average covers midwestern Brazilian areas from 2009 to 2019.

The LST variable had a similar pattern to xCH₄ throughout the time series with the increase starting in the year of 2014 (Fig. 5). We observed lower Rainfall values in 2019. Meanwhile, the spectral index variable of EVI showed outliers for all years throughout the time series (Fig. 5).

Table 1 shows the results of the principal components analysis, with eigenvalues and correlation coefficients. The first principal component (PC1) explained 43.42% of the total variability, meanwhile, the second principal component (PC2) explained 35.02, totalling 78.43% of the variability contained in the original data. The screeplot (Fig. 6) corroborated the explanation and formation of two principal components because an observed eigenvalue remained above one unit and thus explained 78.43% of the variability contained in the original data.

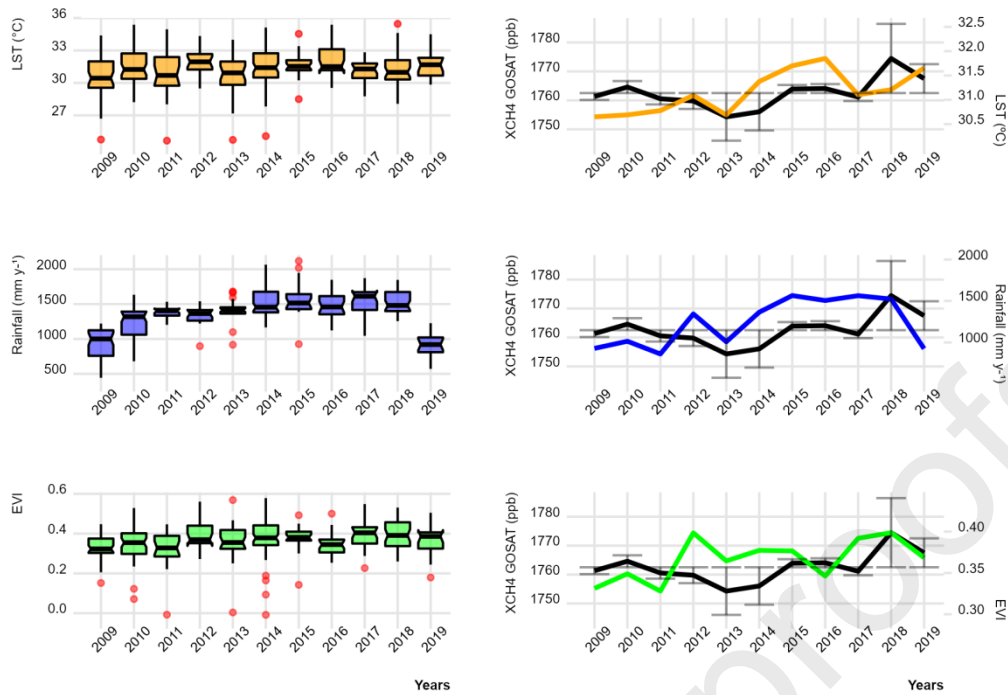


Fig. 5. Annual variations of Land Surface Temperature (LST °C), Rainfall (mm y⁻¹), Enhanced Vegetation Index (EVI) and xCH₄ GOSAT variables cover midwestern Brazilian areas 2009–2019. Outliers beyond the range are denoted by red color.

Fig. 7 presents the resulting biplot of the principal components analysis for the observed variables over the time series from 2009 to 2019. The LST and xCH₄ vectors are close and in the same quadrant, likewise, the rainfall and EVI vectors. These relations are justified by the positive correlations between these variables. Importantly, the LST and xCH₄ vectors were strongly correlated with the years 2018 and 2019, respectively.

Table 1

Eigenvalues, amount of explained variance, correlation coefficients, and eigenvectors between the CH₄ (xCH₄), the variables LST, EVI, Rainfall, and the total cattle herd (Cattle Heads) are the three first principal components.

Components	PC1	PC2
Eigenvalue	2.17	1.75
Explained variance (%)	43.42	35.02
Cumulative variance (%)	43.42	78.44
Correlation (eigenvector)		

xCH ₄	<u>-0.75</u> (-25.77)	0.40 (9.29)
Rainfall	<u>0.66</u> (20.09)	0.70 (27.78)
LST	<u>-0.67</u> (-20.51)	0.69 (27.40)
EVI	<u>0.65</u> (19.63)	0.72 (30.07)
Cattle Heads	<u>0.55</u> (13.99)	-0.30 (-5.44)
Interpretation	xCH ₄ and LST in contrast with Rainfall, EVI, and Cattle Heads	

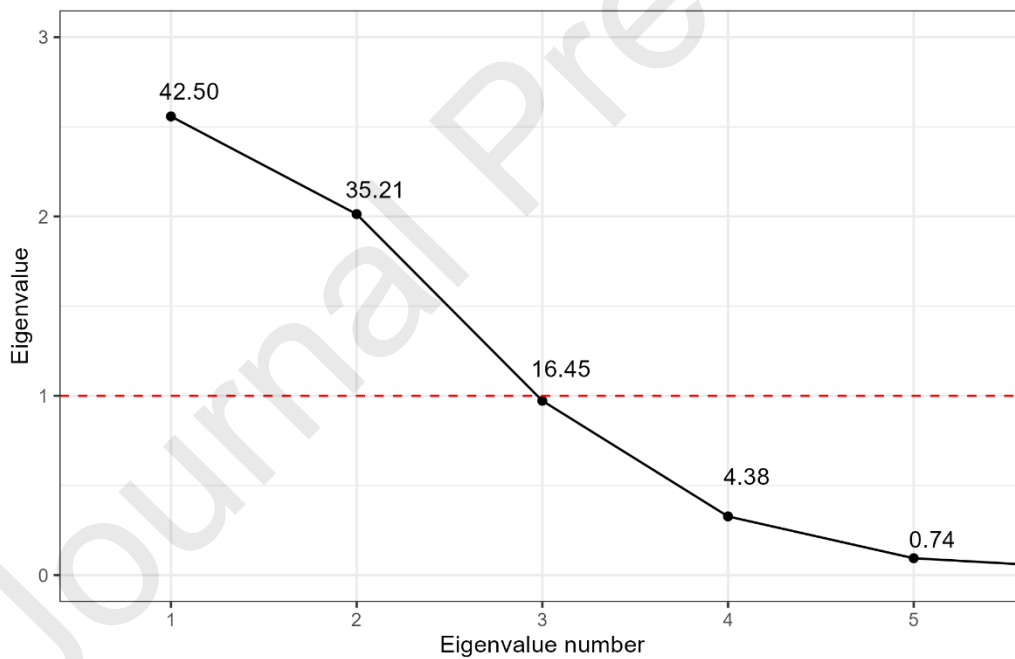


Fig. 6. Screeplot of the principal components analysis, indicating the eigenvalues that are higher than the unity, used for interpreting the results.

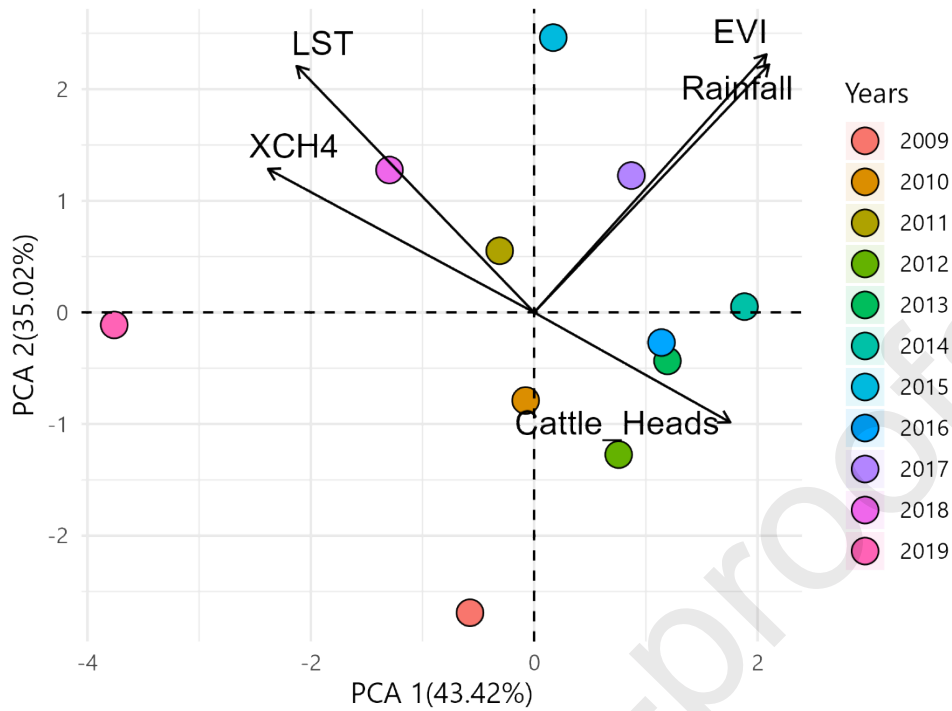


Fig. 7. Biplot of the first two principal components, PC1, and PC2, from the principal component analysis for all soil samples. The variables were the column-averaged dry-air mole fraction of CH₄ (xCH₄), Land Surface Temperature (LST), Enhanced Vegetation Index (EVI), Rainfall, and total cattle herd data (Cattle Heads).

It is worth noting that there was a significant linear effect between xCH₄ and LST ($xCH_4 = 1651.82 + 3.57 \times LST^{\circ}C$) with $P = 2.25e-25$ and with adjusted $R^2 = 0.28$. Our results suggest that the CH₄ concentration will increase around 3.5 ppb per 1 °C (Fig. 8). The analysis of standardized residuals provided noticeable normality in the distribution of 95% of the residuals in a range of -2 and +2. Thus, it evidenced the assumption that the errors are approximately normally distributed with constant variance (Fig S1).

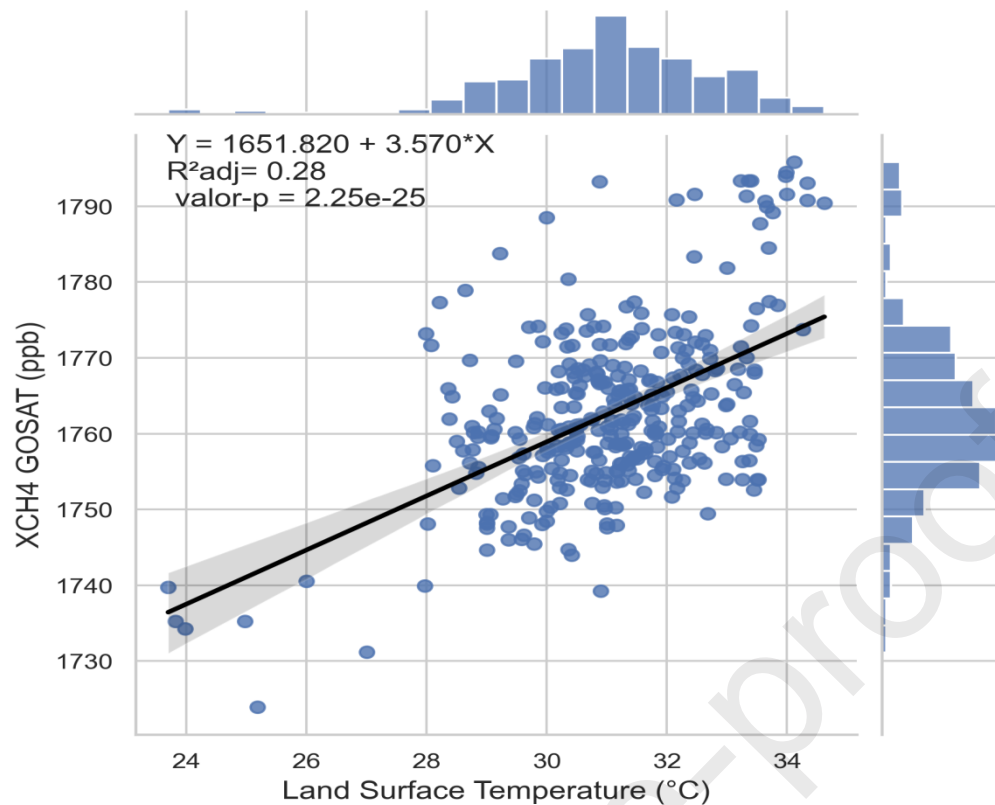


Fig. 8. Linear regression between data monthly averages of Land Surface Temperature (LST °C) and xCH₄ GOSAT (ppb) variables.

4. Discussion

The LST variable demonstrated a strong positive contribution ($p > 0.89$) to xCH₄ emissions over the analyzed time series. Previous studies conducted in China and Mongolia (Dangal et al., 2017) and the United States and Canada (Javadinejad et al., 2019) corroborate these findings. They identified that LST is a significant driver of fluctuations in CH₄ concentrations. Consequently, temperature is recognized as a critical predictor for estimating CH₄ emissions (Javadinejad et al., 2019; Zhang et al., 2017).

As illustrated in Fig. 2, the highest average xCH₄ concentrations occurred between December and February, while the lowest values were observed between May and July. This pattern aligns with the seasonal influence of climate on atmospheric CH₄ levels, reflecting temperature trends in Mato Grosso do Sul (Abreu et al., 2020; Alvares et al., 2013; Ivasko Júnior et al., 2020).

The moderate similarity observed between LST and the Enhanced Vegetation Index (EVI) (Fig. 3) is attributed to anthropogenic heat release due to energy consumption, which reduces vegetation cover (Abdulmana et al., 2021; Kumar et al., 2012). Elevated temperatures are also linked to a reduction in pasture carrying capacity by up to 25%, particularly in central-western Brazil (McManus et al., 2016). Furthermore, interannual drivers such as floods in the Pantanal biome were previously identified as influencing xCH₄ levels. However we did not analyzed the influence of flooding

Global CH₄ emissions from livestock were estimated at 139 Tg in 2019, based on data from GOSAT and TROPOMI satellites (Qu et al., 2021). In the same year, Brazil

contributed approximately 33 Tg of anthropogenic CH₄ emissions, with 70% (23 Tg) originating from enteric fermentation (Maasakkers et al., 2019; Zhang et al., 2021). Mato Grosso do Sul ranked fifth among Brazilian states in livestock emissions, hosting 19.4 million head of cattle in 2019 (IBGE, 2020).

Interestingly, the study identified a weak correlation between cattle herd size and xCH₄ levels. Despite a reduction in cattle numbers, xCH₄ emissions increased, potentially due to observational uncertainties (Kumari et al., 2020) or the mixing of surface CH₄ emissions throughout the atmospheric column, which is less detectable by ground-based measurements such as those from the Total Carbon Column Observation Network (TCCON; Tunncliffe et al., 2020).

A strong positive correlation was observed between EVI and rainfall (Fig. 3), attributed to the impact of precipitation on vegetation vigour and development. Greater rainfall enhances EVI values, which in turn supports higher cattle carrying capacities (Valle Júnior et al., 2020; Rusdy, 2021). This relationship underscores the importance of precipitation in sustaining productive ecosystems in Mato Grosso do Sul.

PC1 made it possible to infer in order of importance: that the xCH₄ variable (-0.75) and the LST variable (-0.67) correlated negatively and the Rainfall (0.66), EVI (0.65), and Cattle Heads (0.55) variables correlated positively (Table 1 and Fig. 7). Tunncliffe et al. (2020) pointed out temperature as the most important drive for CH₄ emissions.

The first axis of the PCA explained 43.42 % of the variability of the vectors analyzed and provided the formation of two distinct groups (Table 1 and Fig. 7). Group I had the biophysical variable “LST” grouped with “XCH₄”, mainly for the year 2019. It is worth noting that group I (LST and xCH₄) was negatively correlated with the other variables. Group II there was a grouping of the biophysical variables of “EVI” and “Rainfall” with “Cattle Heads”, mainly for the year 2013. In axis 2 of the PCA the vectors of “LST”, “XCH₄”, “EVI” and “Rainfall” correlated positively, especially for the year 2019, the “Cattle Heads” vector, was negatively correlated with other variables (Fig. 7).

The positive correlation between the vector of xCH₄ and LST (Fig. 2), is given by the increment in soil respiration due to rising temperatures. A study conducted in Western Amazonia (Zhang et al., 2018) and another investigating CH₄ emission in Brazil from 2010 to 2018 (Tunncliffe et al., 2020) also observed a positive correlation between CH₄ emissions and rising surface temperatures.

Another important factor is that the ENSO phenomenon contributes positively to the warming of extreme wet and dry areas, consequently providing higher global CH₄ emissions (Zhang et al., 2017). Thus, temperature becomes the key variable for understanding the increasing trend in CH₄ emissions over time.

The strong positive correlation between xCH₄ and LST (Fig. 2) highlights the role of rising temperatures in enhancing soil respiration and CH₄ emissions. Studies in the Western Amazon (Zhang et al., 2018) and across Brazil (2010–2018) (Tunncliffe et al., 2020) similarly identified temperature as a critical factor in CH₄ dynamics. The El Niño–Southern Oscillation (ENSO) further exacerbates these emissions by intensifying warming in extremely wet and dry regions (Zhang et al., 2017). An equation was developed to estimate CH₄ emissions based on temperature changes: $xCH_4 = 1651.82 +$

3.57xLST(°C), this equation predicts a 7.14 ppb increase in xCH₄ emissions for every 2°C rise in temperature, emphasizing temperature's pivotal role in CH₄ emission trends.

Monthly xCH₄ levels exhibited a consistent upward trend in April, coinciding with the transition from the rainy to the dry season. Rising temperatures during this period amplify CH₄ emissions. Anthropogenic activities such as land-use change, biomass burning, and fertilizer application also negatively contribute to xCH₄ increases (Singh et al., 2021). Additionally, significant correlations were observed between LST and xCH₄ concentrations in the Amazon biome, highlighting regional variations in CH₄ dynamics (Maria et al., 2023).

5. Conclusion

The analysis of xCH₄ annual averages from 2009 to 2019 revealed no significant upward or downward trend. However, a consistent monotonic increase in the monthly averages for April, particularly after 2014, suggests the influence of seasonal factors on methane dynamics. This pattern highlights the need to consider the seasonal variations in CH₄ concentrations when evaluating long-term trends.

The weak correlation observed between cattle headcounts and xCH₄ emissions was unexpected, suggesting that further research is required, particularly studies integrating terrestrial monitoring of livestock activity. Such investigations will be crucial for better understanding the contributions of livestock to methane emissions in the region.

The LST variable, on the other hand, exhibited a strong positive relationship with xCH₄ concentrations over the 11 years. For every 1°C increase in LST, a 3.57 ppb rise in methane concentrations was observed, highlighting the substantial impact of temperature on methane dynamics. This result strongly suggests that ongoing climate change, particularly temperature increases, will enhance the sensitivity of CH₄ emissions in the central-western region of Brazil, further exacerbating the region's role as a source of greenhouse gases.

Finally, anthropogenic activities such as land-use change, fertilizer application, and biomass burning will likely contribute to increasing xCH₄ concentrations. While no direct relationship was found between cattle herds and xCH₄ emissions, it is essential to address the broader factors driving methane emissions. Achieving net-zero emissions and implementing effective mitigation strategies is urgent for ensuring the long-term sustainability of both the environment and the economy in the region.

Acknowledgments

The study was funded by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brazil (CAPES). I thank the group Geotecnologia Aplicada em Agricultura e Floresta (GAAF), SojaMaps and the Mato Grosso State University (UNEMAT) for the infrastructure of this study.

References

Abdulmana, S., Lim, A., Wongsai, S., Wongsai, N., 2021. Land surface temperature and vegetation cover changes and their relationships in Taiwan from 2000 to 2020.

- Remote Sens. Appl. Soc. Environ. 24, 100636.
<https://doi.org/10.1016/j.rsase.2021.100636>
- Abreu, M.; Lins, T.M.P.; Oliveira-Junior, J.F.; Oliveira, S.S.; Fernandes, W.; Alameida, L.T., 2020. COMPARISON AND VALIDATION OF TRMM SATELLITE PRECIPITATION ESTIMATES AND DATA OBSERVED IN MATO GROSSO DO SUL STATE, BRAZIL. *Rev. Bras. Climatol.* 27.
<https://doi.org/10.5380/abclima.v27i0.68569>
- Alvares, C.A., Stape, J.L., Sentelhas, P.C., de Moraes Gonçalves, J.L., Sparovek, G., 2013. Köppen's climate classification map for Brazil. *Meteorol. Zeitschrift* 22, 711–728. <https://doi.org/10.1127/0941-2948/2013/0507>
- Bansal, S., Katyal, D., Saluja, R., Chakraborty, M., Garg, J.K., 2018. Remotely sensed MODIS wetland components for assessing the variability of CH₄ emissions in Indian tropical/subtropical wetlands. *Int. J. Appl. Earth Obs. Geoinf.* 64, 156–170.
<https://doi.org/10.1016/j.jag.2017.08.011>
- Barbosa, M.L.F., Delgado, R.C., Teodoro, P.E., Pereira, M.G., Correia, T.P., de Mendonça, B.A.F., Ávila Rodrigues, R. de, 2019. Occurrence of fire foci under different land uses in the State of Amazonas during the 2005 drought. *Environ. Dev. Sustain.* 21, 2707–2720. <https://doi.org/10.1007/s10668-018-0157-4>
- Beck, H.E., Zimmermann, N.E., McVicar, T.R., Vergopolan, N., Berg, A., Wood, E.F., 2018. Present and future köppen-geiger climate classification maps at 1-km resolution. *Sci. Data* 5, 1–12. <https://doi.org/10.1038/sdata.2018.214>
- Bergier, I., Silva, A.P.S., Abreu, U.G.P. de, Oliveira, L.O.F. de, Tomazi, M., Dias, F.R.T., Urbanetz, C., Nogueira, É., Borges-Silva, J.C., 2019a. Could bovine livestock intensification in Pantanal be neutral regarding enteric CH₄ emissions? *Sci. Total Environ.* 655, 463–472. <https://doi.org/10.1016/j.scitotenv.2018.11.178>
- Bergier, I., Silva, A.P.S., Abreu, U.G.P. de, Oliveira, L.O.F. de, Tomazi, M., Dias, F.R.T., Urbanetz, C., Nogueira, É., Borges-Silva, J.C., 2019b. Could bovine livestock intensification in Pantanal be neutral regarding enteric CH₄ emissions? *Sci. Total Environ.* 655, 463–472. <https://doi.org/10.1016/j.scitotenv.2018.11.178>
- Bungenstab, D.J., Gomes, R.C., 2016. Changes in cattle herd composition and its implications on greenhouse gases emissions in Mato Grosso do Sul State between 2010 and 2014. *Embrapa Gado*
- Chen, Z., Liu, J., Henze, D.K., Huntzinger, D.N., Wells, K.C., Sitch, S., Friedlingstein, P., Joetzjer, E., Bastrikov, V., Goll, D.S., Haverd, V., Jain, A.K., Kato, E., Lienert, S., Lombardozzi, D.L., McGuire, P.C., Melton, J.R., Nabel, J.E.M.S., Poulter, B., Tian, H., Wiltshire, A.J., Zaehle, S., Miller, S.M., 2021. Linking global terrestrial CO₂ fluxes and environmental drivers: Inferences from the Orbiting Carbon Observatory 2 satellite and terrestrial biospheric models. *Atmos. Chem. Phys.* 21, 6663–6680. <https://doi.org/10.5194/acp-21-6663-2021>
- Chevallier, F., Feng, L., Bösch, H., Palmer, P.I., Rayner, P.J., 2010. On the impact of transport model errors for the estimation of CO₂ surface fluxes from GOSAT observations. *Geophys. Res. Lett.* 37.

<https://doi.org/https://doi.org/10.1029/2010GL044652>

- Crippa, M., Solazzo, E., Huang, G., Guizzardi, D., Koffi, E., Muntean, M., Schieberle, C., Friedrich, R., Janssens-Maenhout, G., 2020. High resolution temporal profiles in the Emissions Database for Global Atmospheric Research. *Sci. Data* 7. <https://doi.org/10.1038/s41597-020-0462-2>
- Dangal, S.R.S., Tian, H., Zhang, B., Pan, S., Lu, C., Yang, J., 2017. CH₄ emission from global livestock sector during 1890-2014: Magnitude, trends and spatiotemporal patterns. *Glob. Chang. Biol.* 23, 4147–4161. <https://doi.org/10.1111/gcb.13709>
- Deng, J. 1982. The grey control system. *Journal of Huazhong Institute of Technology*, 10 (3) (1982), pp. 9-18
- De Lima, R.A.F., Oliveira, A.A., Pitta, G.R., de Gasper, A.L., Vibrans, A.C., Chave, J., ter Steege, H., Prado, P.I., 2020. The erosion of biodiversity and biomass in the Atlantic Forest biodiversity hotspot. *Nat. Commun.* 11. <https://doi.org/10.1038/s41467-020-20217-w>
- Funk, C., Peterson, P., Landsfeld, M., Pedreros, D., Verdin, J., Shukla, S., Husak, G., Rowland, J., Harrison, L., Hoell, A., Michaelsen, J., 2015. The climate hazards infrared precipitation with stations—a new environmental record for monitoring extremes. *Sci. Data* 2, 150066. <https://doi.org/10.1038/sdata.2015.66>
- Gomes, L.; Simões, S.J.C.; Dalla Nora, E.L.; de Sousa-Neto, E.R.; Forti, M.C.; Ometto, J.P.H.B. 2019. Agricultural Expansion in the Brazilian Cerrado: Increased Soil and Nutrient Losses and Decreased Agricultural Productivity. *Land*, 8, 12. <https://doi.org/10.3390/land8010012>
- Gujarati DN, Porter DC. 2011. *Econometria básica-5*. Amgh Editora.
- Gulev, S.K., Thorne, P.W., Ahn, J., Dentener, F.J., Domingues, C.M., Gerland, S. 2023 Changing state of the climate system Change IPoC (Ed.), *Climate Change 2021 – The Physical Science Basis: Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, Cambridge University Press, Cambridge (2023), pp. 287-422. <https://www.cambridge.org/core/product/8B8FB442BA38A2C314ADD4136A9FE2E8>
- Huete, A., Didan, K., Miura, T., Rodriguez, E.P., Gao, X., Ferreira, L.G., 2002. Overview of the radiometric and biophysical performance of the MODIS vegetation indices. *Remote Sens. Environ.* 83, 195–213. [https://doi.org/https://doi.org/10.1016/S0034-4257\(02\)00096-2](https://doi.org/https://doi.org/10.1016/S0034-4257(02)00096-2)
- Huete, A.R., Liu, H.Q., Batchily, K., van Leeuwen, W., 1997. A comparison of vegetation indices over a global set of TM images for EOS-MODIS. *Remote Sens. Environ.* 59, 440–451. [https://doi.org/https://doi.org/10.1016/S0034-4257\(96\)00112-5](https://doi.org/https://doi.org/10.1016/S0034-4257(96)00112-5)
- IBGE - Instituto Brasileiro de Geografia e Estatística., 2020. *Conheca cidades e Estados do Brasil*. [WWW Document]. 2020.

- IPCC, 2021. Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Masson-Delmotte, V., P. Zhai, A. Pirani, S.L. Connors, C. Péan, S. Berger, N. Caud, Y. Chen, [WWW Document]]. 2021. URL <https://www.ipcc.ch/report/2019-refinement-to-the-2006-ipcc-guidelines-for-national-greenhouse-gas-inventories/> (accessed 8.31.21).
- Ivasko Júnior, S., Mastella, A.D.F., Tres, A., Tetto, A.F., Wendling, W.T., Soares, R.V., 2020. Classificação Do Estado De Mato Grosso Do Sul Segundo Sistema De Zonas De Vida De Holdridge. *Rev. Bras. Climatol.* 26, 629–645. <https://doi.org/10.5380/abclima.v26i0.61843>
- Javadinejad, S., Eslamian, S., Ostad-Ali-Askari, K., 2019. Investigation of monthly and seasonal changes of CH₄ gas with respect to climate change using satellite data. *Appl. Water Sci.* 9. <https://doi.org/10.1007/s13201-019-1067-9>
- Kendall, M.G., 1948. Rank correlation methods., Rank correlation methods. Griffin, Oxford, England.
- Kenea, S.-T., Lee, H.-Y., Joo, S.-W., Li, S.-L., Labzovskii, L.D., Chung, C.-Y., Kim, Y.-H., 2021. Interannual Variability of Atmospheric CH₄ and Its Driver Over South Korea Captured by Integrated Data in 2019. *Remote Sens.* 13. <https://doi.org/10.3390/rs13122266>
- Kumar, K.S., Bhaskar, P.U., Padmakumari, K., 2012. Estimation of Land Surface Temperature To Study Urban Heat Island Effect Using Landsat Etm+ Image. *Int. J. Eng. Sci. Technol.* 4, 771–778.
- Kumari, S., Fagodiya, R.K., Hiloidhari, M., Dahiya, R.P., Kumar, A., 2020. CH₄ production and estimation from livestock husbandry: A mechanistic understanding and emerging mitigation options. *Sci. Total Environ.* 709, 136135. <https://doi.org/10.1016/j.scitotenv.2019.136135>
- Kuze, A., Suto, H., Nakajima, M., Hamazaki, T., 2009. Thermal and near infrared sensor for carbon observation Fourier-transform spectrometer on the Greenhouse Gases Observing Satellite for greenhouse gases monitoring. *Appl. Opt.* 48, 6716. <https://doi.org/10.1364/AO.48.006716>
- Lan, X., Tans, P., Thoning, K.W. 2023. National Oceanic and Atmospheric Administration GMLNG (Ed.), Trends in Globally-averaged CO₂ Determined From NOAA Global Monitoring Laboratory Measurements (2023) <https://gml.noaa.gov/ccgg/trends/global.html>, Accessed 12th Nov 2024
- Maria, L. S., Rossi, F.S., Costa, L.C., Campos, M.O., Blas, J.C.G., Panosso, A.P., Silva, J.I.D., Silva Junior, C.A., La Scala Jr., N. 2023. Spatiotemporal analysis of atmospheric XCH₄ as related to fires in the Amazon biome during 2015–2020. *Remote Sensing Applications: Society and Environment*. Volume 30. <https://doi.org/10.1016/j.rsase.2023.100967>
- Marengo, J.A., Cunha, A.P., Cuartas, L.A., Deusdará Leal, K.R., Broedel, E., Seluchi,

- M.E., Michelin, C.M., De Praga Baião, C.F., Chuchón Ângulo, E., Almeida, E.K., Kazmierczak, M.L., Mateus, N.P.A., Silva, R.C. Bender, F. 2021. Extreme Drought in the Brazilian Pantanal in 2019–2020: Characterization, Causes, and Impacts. *Front. Water*, 3:639204. doi: 10.3389/frwa.2021.639204
- McManus, C., Barcellos, J.O.J., Formenton, B.K., Hermuche, P.M., Carvalho Jr, O.A. de, Guimarães, R., Gianezini, M., Dias, E.A., Lampert, V. do N., Zago, D., Neto, J.B., 2016. Dynamics of Cattle Production in Brazil. *PLoS One* 11, 1–15. <https://doi.org/10.1371/journal.pone.0147138>
- Mendonça, A.K. de S., da Silva, S.A., Pereira, L.Z., Bornia, A.C., de Andrade, D.F., 2020. An overview of environmental policies for mitigation and adaptation to climate change and application of multilevel regression analysis to investigate the CO₂ emissions over the years of 1970 to 2018 in all Brazilian states. *Sustain.* 12, 1–18. <https://doi.org/10.3390/su12219175>
- Morellato, L.P.C., Haddad, C.F.B., 2000. Introduction: The Brazilian Atlantic Forest. *Biotropica* 32, 786–792. [https://doi.org/https://doi.org/10.1111/j.1744-7429.2000.tb00618.x](https://doi.org/10.1111/j.1744-7429.2000.tb00618.x)
- Mousavi, S.M., Falahatkar, S., 2020. Spatiotemporal distribution patterns of atmospheric CH₄ using GOSAT data in Iran. *Environ. Dev. Sustain.* 22, 4191–4207. <https://doi.org/10.1007/s10668-019-00378-5>
- Mousavi, S.M., Falahatkar, S., Farajzadeh, M., 2017. Assessment of seasonal variations of carbon dioxide concentration in Iran using GOSAT data. *Nat. Resour. Forum* 41, 83–91. <https://doi.org/https://doi.org/10.1111/1477-8947.12121>
- Mustafa, F., Wang, H., Bu, L., Wang, Q., Shahzaman, M., Bilal, M., Zhou, M., Iqbal, R., Aslam, R.W., Ali, M.A., Qiu, Z., 2021. Validation of GOSAT and OCO-2 against In Situ Aircraft Measurements and Comparison with CarbonTracker and GEOS-Chem over Qinhuangdao, China. *Remote Sens.* 13. <https://doi.org/10.3390/rs13050899>
- Parker, R.; Boesch, H., 2020. University of Leicester GOSAT Proxy XCH₄ v9.0. Centre for Environmental Data Analysis [WWW Document]. 2020.
- Parker, R., Boesch, H., Cogan, A., Fraser, A., Feng, L., Palmer, P.I., Messerschmidt, J., Deutscher, N., Griffith, D.W.T., Notholt, J., Wennberg, P.O., Wunch, D., 2011. CH₄ observations from the Greenhouse Gases Observing SATellite: Comparison to ground-based TCCON data and model calculations. *Geophys. Res. Lett.* 38. <https://doi.org/10.1029/2011GL047871>
- Parker, R.J., Boesch, H., Byckling, K., Webb, A.J., Palmer, P.I., Feng, L., Bergamaschi, P., Chevallier, F., Notholt, J., Deutscher, N., Warneke, T., Hase, F., Susmann, R., Kawakami, S., Kivi, R., Griffith, D.W.T., Velasco, V., 2015. Assessing 5 years of GOSAT Proxy XCH₄ data and associated uncertainties. *Atmos. Meas. Tech.* 8, 4785–4801. <https://doi.org/10.5194/amt-8-4785-2015>
- Parker, R.J., Boesch, H., Wooster, M.J., Moore, D.P., Webb, A.J., Gaveau, D., Murdiyarso, D., 2016. Atmospheric

- CH₄ and CO₂ enhancements and biomass burning emission ratios derived from satellite observations of the 2015 Indonesian fire plumes. *Atmos. Chem. Phys.* 16, 10111–10131. <https://doi.org/10.5194/acp-16-10111-2016>
- Parker, R.J., Webb, A., Boesch, H., Somkuti, P., Barrio Guillo, R., Di Noia, A., Kalaitzi, N., Anand, J.S., Bergamaschi, P., Chevallier, F., Palmer, P.I., Feng, L., Deutscher, N.M., Feist, D.G., Griffith, D.W.T., Hase, F., Kivi, R., Morino, I., Notholt, J., Oh, Y.-S., Ohyama, H., Petri, C., Pollard, D.F., Roehl, C., Sha, M.K., Shiomi, K., Strong, K., Sussmann, R., Té, Y., Velazco, V.A., Warneke, T., Wennberg, P.O., Wunch, D., 2020. A decade of GOSAT Proxy satellite {CH_4} observations. *Earth Syst. Sci. Data* 12, 3383–3412. <https://doi.org/10.5194/essd-12-3383-2020>
- Pettitt, A.N., 1979. A Non-Parametric Approach to the Change-Point Problem
Published by : Wiley for the Royal Statistical Society A Non-parametric Approach to the Change-point Problem. *Journal of the Royal Statistical Society. Series C (Applied Statistics)* 28, 126–135.
- Peters, W., Jacobson, A.R., Sweeney, C., Andrews, A.E., Conway, T.J., Masarie, K., Miller, J.B., Bruhwiler, L.M.P., Pétron, G., Hirsch, A.I., Worthy, D.E.J., van der Werf, G.R., Randerson, J.T., Wennberg, P.O., Krol, M.C., Tans, P.P., 2007. An atmospheric perspective on North American carbon dioxide exchange: CarbonTracker. *Proc. Natl. Acad. Sci.* 104, 18925–18930. <https://doi.org/10.1073/pnas.0708986104>
- R Development Core Team, 2017. A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria.
- Ross, A.N., Wooster, M.J., Boesch, H., Parker, R., 2013. First satellite measurements of carbon dioxide and CH₄ emission ratios in wildfire plumes. *Geophys. Res. Lett.* 40, 4098–4102. <https://doi.org/10.1002/grl.50733>
- Rusdy, M., 2021. Grass-legume intercropping for sustainability animal production in the tropics. *CAB Rev. Perspect. Agric. Vet. Sci. Nutr. Nat. Resour.* 16. <https://doi.org/10.1079/PAVSNNR202116021>
- Saunois, M., R. Stavert, A., Poulter, B., Bousquet, P., G. Canadell, J., B. Jackson, R., A. Raymond, P., J. Dlugokencky, E., Houweling, S., K. Patra, P., Ciais, P., K. Arora, V., Bastviken, D., Bergamaschi, P., R. Blake, D., Brailsford, G., Bruhwiler, L., M. Carlson, K., Carrol, M., Castaldi, S., Chandra, N., Crevoisier, C., M. Crill, P., Covey, K., L. Curry, C., Etiope, G., Frankenberg, C., Gedney, N., I. Hegglin, M., Höglund-Isaksson, L., Hugelius, G., Ishizawa, M., Ito, A., Janssens-Maenhout, G., M. Jensen, K., Joos, F., Kleinen, T., B. Krummel, P., L. Langenfelds, R., G. Laruelle, G., Liu, L., MacHida, T., Maksyutov, S., C. McDonald, K., McNorton, J., A. Miller, P., R. Melton, J., Morino, I., Müller, J., Murguía-Flores, F., Naik, V., Niwa, Y., Noce, S., O'Doherty, S., J. Parker, R., Peng, C., Peng, S., P. Peters, G., Prigent, C., Prinn, R., Ramonet, M., Regnier, P., J. Riley, W., A. Rosentreter, J., Segers, A., J. Simpson, I., Shi, H., J. Smith, S., Paul Steele, L., F. Thornton, B., Tian, H., Tohjima, Y., N. Tubiello, F., Tsuruta, A., Viovy, N., Voulgarakis, A., S.

- Weber, T., Van Weele, M., R. Van Der Werf, G., F. Weiss, R., Worthy, D., Wunch, D., Yin, Y., Yoshida, Y., Zhang, W., Zhang, Z., Zhao, Y., Zheng, B., Zhu, Qing, Zhu, Qian, Zhuang, Q., 2020. The global CH₄ budget 2000-2017. *Earth Syst. Sci. Data* 12, 1561–1623. <https://doi.org/10.5194/essd-12-1561-2020>
- SEEG – Greenhouse Gas Emissions and Removals Estimation System 2020. Climate Observatory. Available at: <<http://seeg.eco.br/>>. Accessed on: July 29/2023.
- Singh, A., Kuttippurath, J., Abbhishek, K., Mallick, N., Raj, S., Chander, G., & Dixit, S. 2021. Biogenic link to the recent increase in atmospheric methane over India. *Journal of Environmental Management*, 289, 112526. <https://doi.org/10.1016/j.jenvman.2021.112526>
- Teodoro, P.E., de Oliveira-Júnior, J.F., da Cunha, E.R., Correa, C.C.G., Torres, F.E., Bacani, V.M., Gois, G., Ribeiro, L.P., 2016. Cluster analysis applied to the spatial and temporal variability of monthly rainfall in Mato Grosso do Sul State, Brazil. *Meteorol. Atmos. Phys.* 128, 197–209. <https://doi.org/10.1007/s00703-015-0408-y>
- Teodoro, P.E.; Maria, L.d.S.; Rodrigues, J.M.A.; Silva, A.d.A.e.; Silva, M.C.M.d.; Souza, S.S.d.; Rossi, F.S.; Teodoro, L.P.R.; Della-Silva, J.L.; Delgado, R.C. 2022. Wildfire Incidence throughout the Brazilian Pantanal Is Driven by Local Climate Rather Than Bovine Stocking Density. *Sustainability*, 14, 10187. <https://doi.org/10.3390/su141610187>
- Tian, H., Chen, G., Lu, C., Xu, X., Ren, W., Zhang, B., Banger, K., Tao, B., Pan, S., Liu, M., Zhang, C., Bruhwiler, L., Wofsy, S., 2015. Global CH₄ and nitrous oxide emissions from terrestrial ecosystems due to multiple environmental changes. *Ecosyst. Heal. Sustain.* 1, 1–20. <https://doi.org/10.1890/EHS14-0015.1>
- Tian, H., Lu, C., Ciais, P., Michalak, A.M., Canadell, J.G., Saikawa, E., Huntzinger, D.N., Gurney, K.R., Sitch, S., Zhang, B., Yang, J., Bousquet, P., Bruhwiler, L., Chen, G., Dlugokencky, E., Friedlingstein, P., Melillo, J., Pan, S., Poulter, B., Prinn, R., Saunois, M., Schwalm, C.R., Wofsy, S.C., 2016. The terrestrial biosphere as a net source of greenhouse gases to the atmosphere. *Nature* 531, 225–228. <https://doi.org/10.1038/nature16946>
- Tunnicliffe, R., L. Ganesan, A., J. Parker, R., Boesch, H., Gedney, N., Poulter, B., Zhang, Z., Walter, D., Rigby, M., Henne, S., Young, D., O'Doherty, S., 2020. Quantifying sources of Brazil's CH₄ emissions between 2010 and 2018 from satellite data. *Atmos. Chem. Phys.* 20, 13041–13067. <https://doi.org/10.5194/acp-20-13041-2020>
- Valle Júnior, L.C.G., Ventura, T.M., Souza, R.S.R., de S. Nogueira, J., de A. Lobo, F., Vourlitis, G.L., Rodrigues, T.R., 2020. Comparative assessment of modelled and empirical reference evapotranspiration methods for a brazilian savanna. *Agric. Water Manag.* 232, 106040. <https://doi.org/10.1016/j.agwat.2020.106040>
- Vermote, E., Justice, C., Claverie, M., Franch, B., 2016. Preliminary analysis of the performance of the Landsat 8/OLI land surface reflectance product. *Remote Sens. Environ.* 185, 46–56. <https://doi.org/10.1016/j.rse.2016.04.008>
- Wan, Z., S. Hook, G.H., 2015. MOD11A2 MODIS/Terra Land Surface

- Temperature/Emissivity 8-Day L3 Global 1km SIN Grid V006. 2015, distributed by NASA EOSDIS Land Processes DAAC [WWW Document]. 2015.
- Wang, G., Cai, W., Santoso, A., 2021. Simulated Thermocline Tilt Over the Tropical Indian Ocean and Its Influence on Future Sea Surface Temperature Variability. *Geophys. Res. Lett.* 48. <https://doi.org/10.1029/2020GL091902>
- Walter, A.; Seabra, J.; Rocha, J.; Guarenghi, M.; Vieira, N.; Damame, D.; Santos, J.L., 2021. Spatially Explicit Assessment of Suitable Conditions for the Sustainable Production of Aviation Fuels in Brazil. *Land*, 10, 705. <https://doi.org/10.3390/land10070705>
- Webb, A.J., Bösch, H., Parker, R.J., Gatti, L. V., Gloor, E., Palmer, P.I., Basso, L.S., Chipperfield, M.P., Correia, C.S.C., Domingues, L.G., Feng, L., Gonzi, S., Miller, J.B., Warneke, T., Wilson, C., 2016. CH₄ concentrations over the Amazon from GOSAT consistent with in situ vertical profile data. *J. Geophys. Res.* 121, 11,006–11,020. <https://doi.org/10.1002/2016JD025263>
- Wilson, C., Chipperfield, M.P., Gloor, M., Parker, R.J., Boesch, H., McNorton, J., Gatti, L. V., Miller, J.B., Basso, L.S., Monks, S.A., 2020. Large and increasing CH₄ emissions from Eastern Amazonia derived from satellite data, 2010--2018. *Atmos. Chem. Phys. Discuss.* 2020, 1–38. <https://doi.org/10.5194/acp-2020-1136>
- Wu, J., Chavana-Bryant, C., Prohaska, N., Serbin, S.P., Guan, K., Albert, L.P., Yang, X., van Leeuwen, W.J.D., Garnello, A.J., Martins, G., Malhi, Y., Gerard, F., Oliveira, R.C., Saleska, S.R., 2017. Convergence in relationships between leaf traits, spectra and age across diverse canopy environments and two contrasting tropical forests. *New Phytol.* 214, 1033–1048. <https://doi.org/10.1111/nph.14051>
- Zhang, Z., Zimmermann, N.E., Calle, L., Hurtt, G., Chatterjee, A., Poulter, B., 2018. Enhanced response of global wetland CH₄ emissions to the 2015-2016 El Niño-Southern Oscillation event. *Environ. Res. Lett.* 13. <https://doi.org/10.1088/1748-9326/aac939>
- Zhang, Z., Zimmermann, N.E., Stenke, A., Li, X., Hodson, E.L., Zhu, G., Huang, C., Poulter, B., 2017. Emerging role of wetland CH₄ emissions in driving 21st century climate change. *Proc. Natl. Acad. Sci. U. S. A.* 114, 9647–9652. <https://doi.org/10.1073/pnas.1618765114>

Supplementary Material

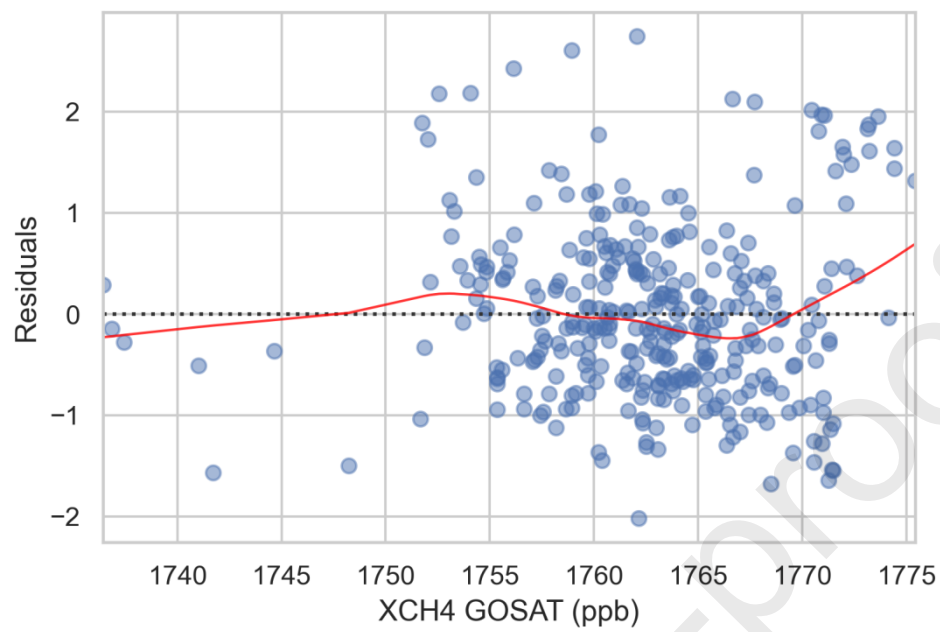


Fig. S1. Plot of residuals versus xCH₄ GOSAT (ppb) level x.

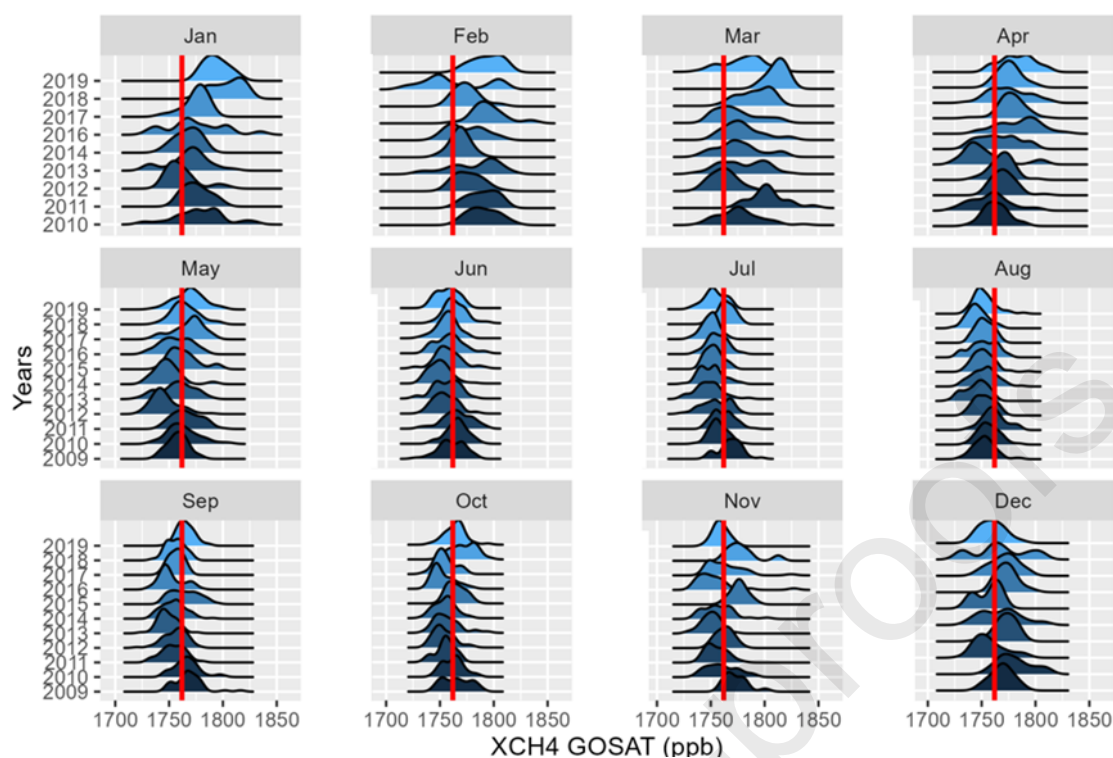


Fig. S2. Mann Kendall test statistics for monthly average xCH₄ GOSAT (ppb). With significant value for April average ($z = 2.0241$ and $P = 0.0429$) and Pettitt's test ($P = 0.048$), with increasing monotonic trend.

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: